Reliable Collective Reasoning and Evaluation for AI Agents

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

Xinya Du, Assistant Professor, Computer Science, The University of Texas at Dallas xinya.du@utdallas.edu https://xinyadu.github.io

Program Officer: Dr. Predrag Neskovic, Code 311 — Mathematical Data Science

Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Machine Learning

Problem. Teams of language-model agents can compare answers, critique reasoning steps, and combine evidence from multiple reports. However, agreement among agents is not necessarily independent confirmation: several judges can inherit one outdated source or reproduce the same mistake. For example, five agents assessing whether a coastal route has reopened may all cite summaries copied from one incorrect bulletin. A simple majority would be confidently wrong. The open question is how to turn heterogeneous, dependent agent judgments into a reliable assessment when labeled examples are scarce.

Central hypothesis. Collective reasoning will be more accurate and better calibrated if aggregation jointly estimates each agent's expertise and how much independent information its judgment contributes. We will connect this uncertainty model to targeted follow-up checks, rather than equating more votes with more evidence.

Research plan.

- **Model dependent judgments and sources.** Represent agent answers and critiques as noisy observations of latent claim correctness. Use shared citations, agreement patterns, and limited reference labels to infer expertise and correlated errors. We will study when reliability and dependence are identifiable, and quantify the effective number of independent assessments.
- **Aggregate reasoning with calibrated uncertainty.** Develop weighting and probabilistic fusion methods that account for task difficulty, correlated mistakes, and conflicting evidence. Instead of ranking agents by average accuracy alone, estimate how much a new judgment changes the collective belief. The system will abstain when apparently unanimous opinions share weak support.
- **Choose informative verification actions.** Formulate the next step as a decision between asking another agent, consulting an independent source, testing an inference, or stopping. We will compare uncertainty reduction and final accuracy against the cost of each action, and study sequential risk control when judgments remain dependent.

Evaluation and expected outcomes. PRD-style judge panels, MEQA and MuSiQue reasoning questions, and controlled duplicate-source cases will test consensus quality. Measures include calibration, error at matched answer coverage, dependence estimation, and verification cost. Baselines include majority voting and independent-expertise weighting.

Naval relevance. In maritime information analysis, reports of the same coastal incident may circulate through multiple channels but originate from a single erroneous observation. An AI team that distinguishes duplicated reporting from independent corroboration can help analysts decide which claims warrant another source check. The central contribution aligns with Mathematical Data Science interests in principled multi-source information integration and decisions under uncertainty.

Prior work and PI qualifications. Our synthetic-error verifier FG-PRM (Li et al., 2025), peer-judgment framework PRD (Li et al., 2024), and hierarchical process rewards for retrieval, HiPRAG (Wu et al., 2026), provide the starting point. The PI holds a 2024 NSF CAREER award and led RESIN's DARPA KAIROS delivery-system integration as a postdoc; the team achieved top performance among participants.